Multimodal Cancer Cell Classification

30 versions, six diagnosed failures, one breakthrough

A methodological run at the Multimodal Cancer Classification challenge — where each diagnosed failure shaped the next version, and the wins came from reading the losses faithfully.

PUBLIC LB

0.8392

v58 co-attention seed2 · #3 on LB

LIFT VS V19

+0.094

VERSIONS

30

BF · 128 × 128

FL · 128 × 128

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Advanced Deep Learning for Image Processing · 1MD042

Uppsala University · June 4, 2026

Binary cell classification on paired BF + FL

Two structural facts shape every method choice that follows.

114,302
TRAIN CELLS · 12 patients

59,040
TEST CELLS · patient-disjoint

38.8%
POS RATE · pos_weight ≈ 1.58

0.8392
BEST LB · v58 co-attention

- — Per-cell prediction: is the cell from a cancer patient? (weak labels — MAC hypothesis, all cells inherit patient diagnosis)
- — Paired inputs: brightfield + fluorescence, 128×128 grayscale
- — 12 training patients · ~10k cells each — small N
- — Strict patient-disjoint test set — OOD generalisation is the dominant failure mode
- — Metric: AUC · 4 submissions/day · public/private LB split

GRADING BASIS (INSTRUCTOR, MAY 14)
“What is done and how that is presented” — methodology over leaderboard score.

Dual EfficientNet-B0 with patient-MIL aux loss

The interventions that matter sit outside the backbone.

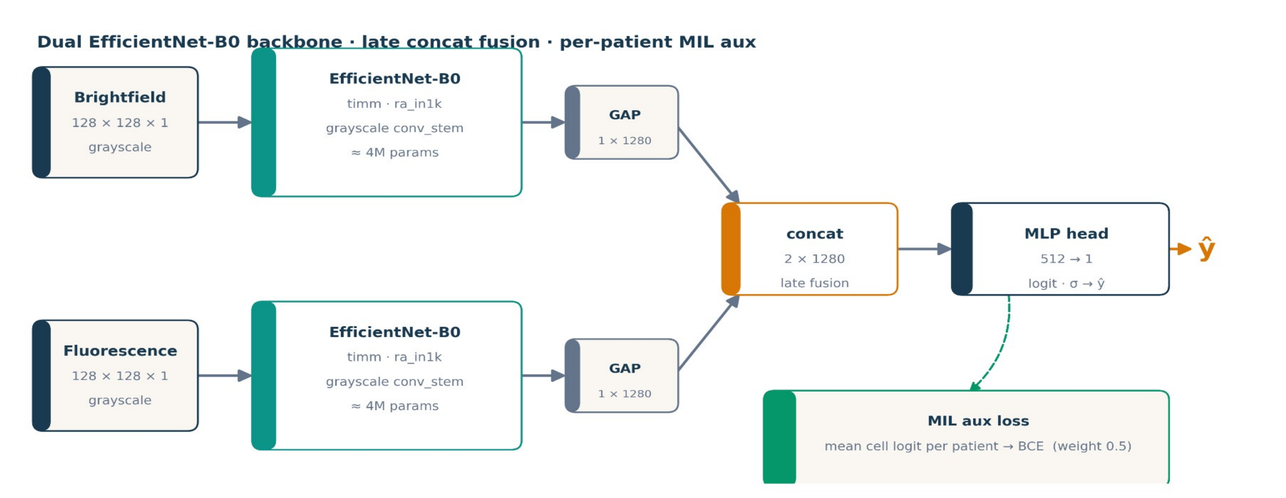

Fig. 1 Backbone is unremarkable on purpose. The interesting parts are at training time (MIL) and test time (AdaBN, stain norm).

- **Late concat fusion** *feature-level after GAP (v38 early fusion regressed −0.031)*
- **Per-patient MIL aux** *mean-pool cell logits per patient, BCE, w=0.5*
- **AdaBN** *refresh BN stats on test set (Li 2016)*
- **Test-set stain norm** *closes the 19.7 % FL mean gap*
- **8-way D4 TTA** *dihedral group, prob-averaged*

30 versions, 5 acts, one breakthrough

Each act answered a question. Each result changed the next question.

ACT 1+2	ACT 3	ACT 4	ACT 5A	ACT 5B
v10–v16	**v17–v19**	**v20–v41**	**v42–v44**	**v46→v47**
Baselines & SSL	*EffNet pivot*	*Regularizer search*	*Hard pseudo-labels*	*Soft distillation*
0.572	**0.7455**	**0.7563**	**0.7812**	**0.8264**
SSL collapse	*supervised floor*	*v41 +0.011*	*v44 +0.025 (Lee 2013)*	*+0.039 + 0.003 · final pick*

HEADLINE *+0.094 LB total — the biggest single jump (+0.039) was soft-pseudo distillation (Hinton 2015).*

v15 / v16: best CV I ever saw, worst LB

CoMIR-style cross-modal SSL — and a 29-point generalisation gap.

What we built

- Two ResNet-18 branches (BF + FL) with projection heads
- NT-Xent contrastive pretrain, τ = 0.1 · 10 epochs
- Supervised fine-tune · 8 epochs · discriminative LR
- **3-fold stratified-group CV** *hid the failure — two of three folds had val patients similar to train*

0.866
CV AUC · 3-fold stratified-group · best I'd ever seen

0.572
PUBLIC LB · v15 submission · 29 AUC points of OOD gap

0.943 → 0.503
v16 RESPONSE · 9 careful fixes · LOPO CV — patient-OOF up, LB still random

DIAGNOSIS

Paired (BF, FL) of cell i always share a patient. Contrastive task solved at ≥95 % by encoding patient identity, not cell content.

v19: drop the SSL, model the test distribution

Five ingredients. None of them are new. Together: +0.174 LB over v15.

0.7455
PUBLIC LB · v19 · supervised floor · held ~3 weeks

+0.174
VS V15 SSL collapse → floor

−0.038
VS prior LB leader · 0.7832 at the time

Five ingredients in v19

— EfficientNet-B0 dual-branch (BF + FL), ImageNet-pretrained, late concat fusion
— Per-patient MIL auxiliary loss — patient-mean-logit BCE, weight 0.5
— AdaBN — refresh BN running stats on the test set (Li 2016)
— Test-set stain normalisation — pixel mean/std from test, not train
— Strong paired augmentation + 8-way D4 group TTA, prob-averaged

TAKE-HOME *The win wasn't a new architecture — it was treating the test distribution as the thing to model.*

v41 vs v43: change one thing, not four

v41 · clean stack on top of v19

0.7563 **+0.011**

- **label smoothing ε = 0.05** *L4 standard*
- **dropout 0.3 → 0.4** *classifier head*
- **paired RRC** *RandomResizedCrop scale 0.85-1.0*
- **multiscale TTA** *3 scales × 8 D4 = 24-way*

v43 · four MORE, stacked at once

0.7444 **−0.012**

- **FL-tuned aug** *CJ 0.5/0.3 + RandomGamma on FL only*
- **WD bump** *1e-4 → 3e-4*
- **3-seed × SWA** *Izmailov 2018 · last 4 epochs*
- **40-way TTA** *5 scales × 8 D4*

RULE → One or two related variables per submitted version. v46 later reverted the FL-tuned aug + WD bump and recovered +0.079.

Use the test set itself as training signal

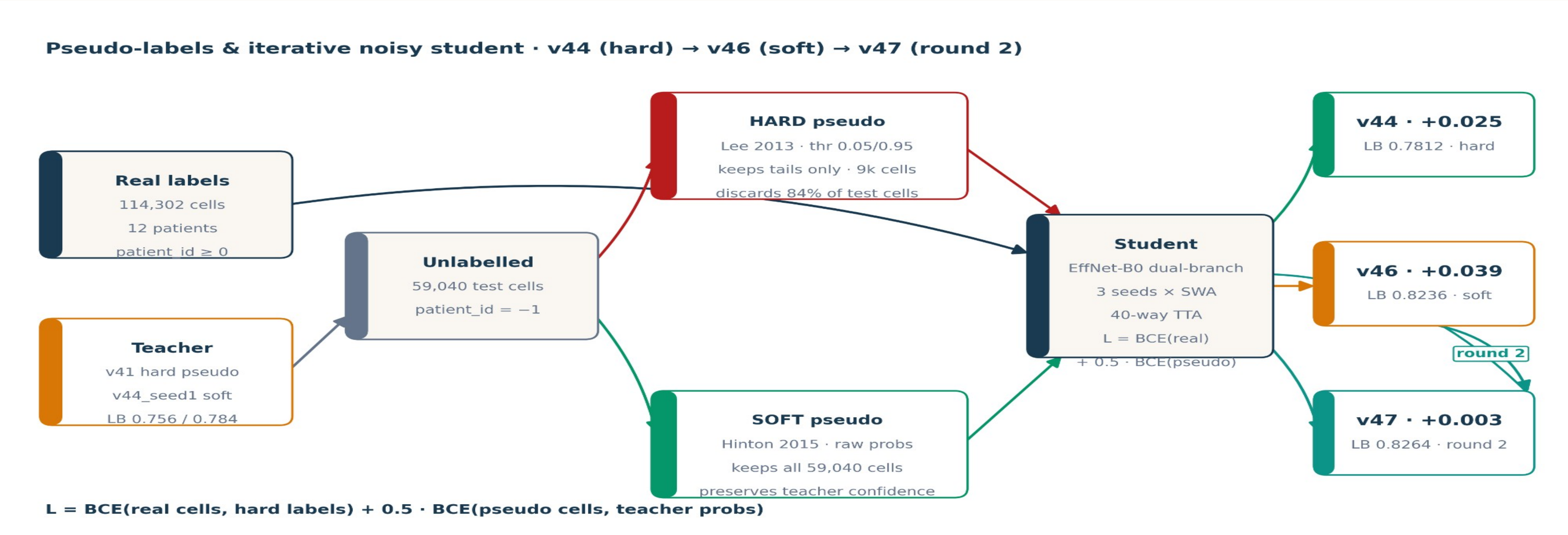

Fig. 2 Same teacher, two ways to use it. Hard discards 84 % of test cells. Soft keeps everything, weighted by teacher confidence. Round 2 uses v46 as the new teacher → v47.

Hard drops 84% of test cells; soft keeps all

Same teacher. Same predictions. Different way of using them.

METHOD	CELLS USED	Δ LB
baseline v41	114,302	—
v44 · hard	+ 9,350 (16 %)	**+0.025**
v46 · soft	+ 59,040 (100 %)	**+0.039**
v47 · round 2	+ 59,040 (100 %)	**+0.003**

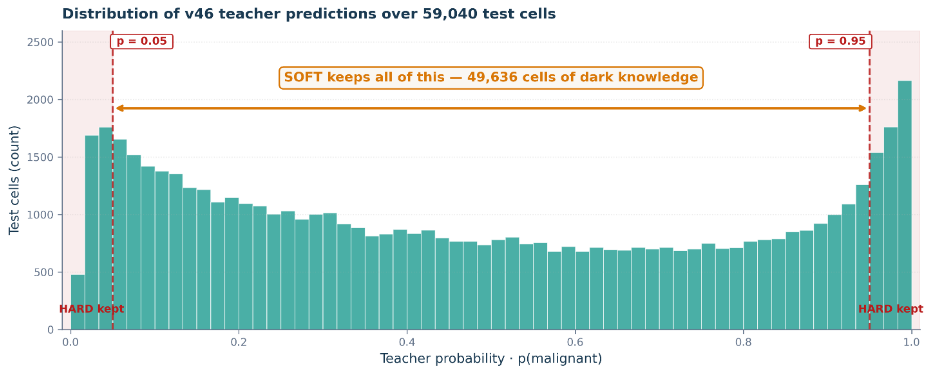

Fig. 3 Distribution of teacher predictions. Red dashed = hard thresholds. Soft keeps every column.

DARK KNOWLEDGE (HINTON 2015)

A cell labelled p = 0.78 by the teacher carries directional + magnitude information that hard binarisation discards. Soft targets preserve all of it.

WHY SOFT ALSO HELPS ROUND 2

v47's +0.003 lift on top of v46's +0.039 is small in mean — but per-seed tr_auc tightened (0.989-0.993 → 0.993-0.994). Round-2 acts as a variance reducer.

The whole story in one chart

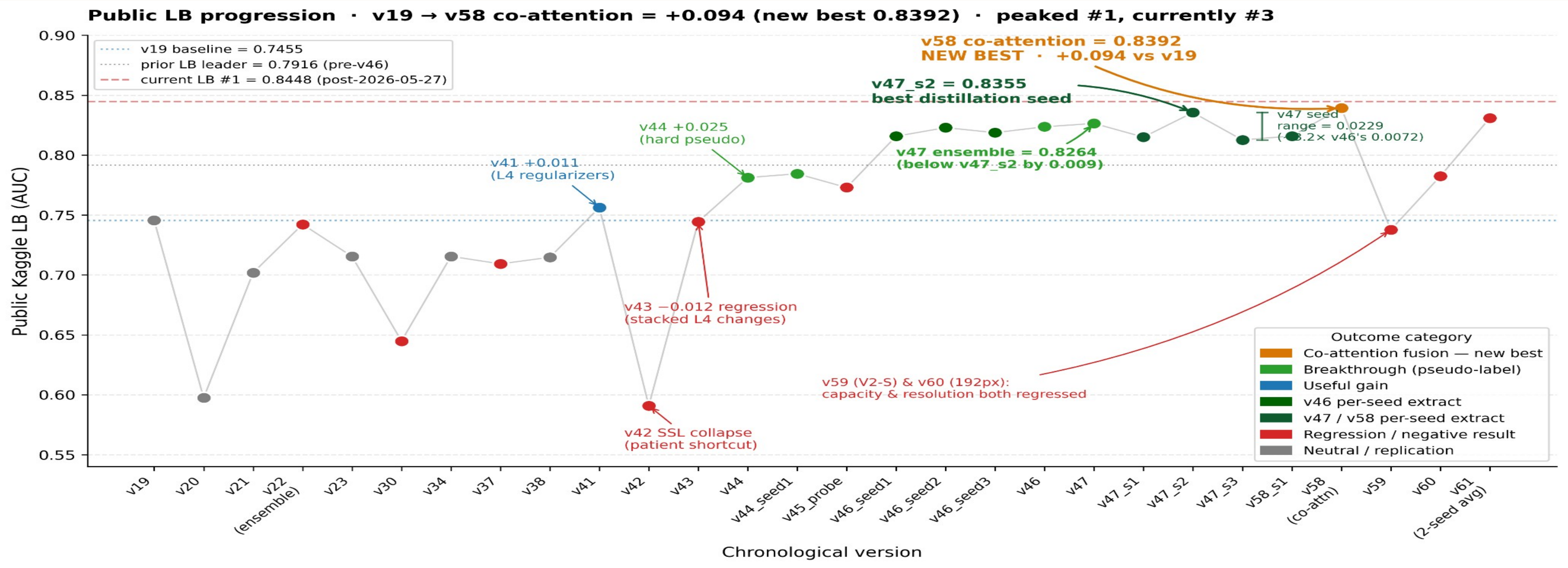

Fig. 4 Public LB across 30+ logged submissions. Amber = co-attention (new best) · green = pseudo-label breakthrough · blue = useful gain · red = regression · grey = neutral. Dashed red = current leader.

Six diagnosed failures that shaped the winning recipe

v22

Cross-recipe ensemble collapse

Sigmoid-avg of v19 + v21 at a 0.05 LB gap regressed below v19 alone (0.7422 vs 0.7455).

RULE → *Don't ensemble across recipes beyond a 0.02 LB gap.*

v27

Held-out val too noisy on N=12

2-patient holdout (pat_5 saturated FL, pat_14 dark FL) selected best_epoch = 0 — untrained.

RULE → *Use public LB as external validator; abandon held-out splits.*

v37 / v38

Lian 2024 transfer attempts

Heavy FL aug + early fusion both regressed (−0.036, −0.031). Lian's 4-ch ≠ our 1-ch grayscale.

RULE → *Re-validate any borrowed ablation on the actual modality.*

v42

SSL patient-shortcut collapse

CoMIR InfoNCE solved by patient-ID (loss = 0.02 in epoch 1). tr_auc 0.96 · LB 0.59.

RULE → *Cross-modal SSL amplifies patient-grouped shortcuts.*

v43

Stacked regulariser regression

Four simultaneous changes on top of v41 → −0.012 LB. No attribution possible.

RULE → *One or two related variables per submission.*

v45_probe

Ensemble rule confirmation

Sigmoid-avg of v41 + v44 (0.025 LB gap) regressed 0.008 below v44. Pearson 0.936.

RULE → *Tightened the rule from §22 to ≤ 0.015 LB gap.*

Every ensemble I tried lost to the best single model

0.8264

3-seed average — BELOW the 0.8355 best single seed

0.7074

orthogonal feature-GBM blend — CV said 0.82, the LB said this

0.7422

cross-recipe average (v22) — below v19 (0.7455) alone

— **Within-recipe: averaging regresses — confirmed on two recipes.** *v47's 3-seed mean (0.8264) fell below its best seed (0.8355); v61's 2-seed co-attention mean (0.8308) fell below its best seed (0.8350). Per-seed LB spans ~0.023 on N=12 — the variance is signal-destroying, not noise to average out.*

— **Cross-model: the stronger member is always pulled down.** *Symmetric averaging drags toward the weaker member, and on this data the quality gap is always wide (v22, v45_probe, and the GBM blend all regressed).*

— **Even an orthogonal model couldn't save it.** *A decorrelated feature-GBM (rank-corr 0.41) looked perfect on paper — but its CV (0.82) did not transfer (LB ~0.60). CV-vs-LB struck the ensemble too.*

TAKE-HOME *On N=12, variance dominates — any averaging regresses below the best single seed. Submit one, never a blend.*

Co-attention broke the plateau — a new best, 0.8392

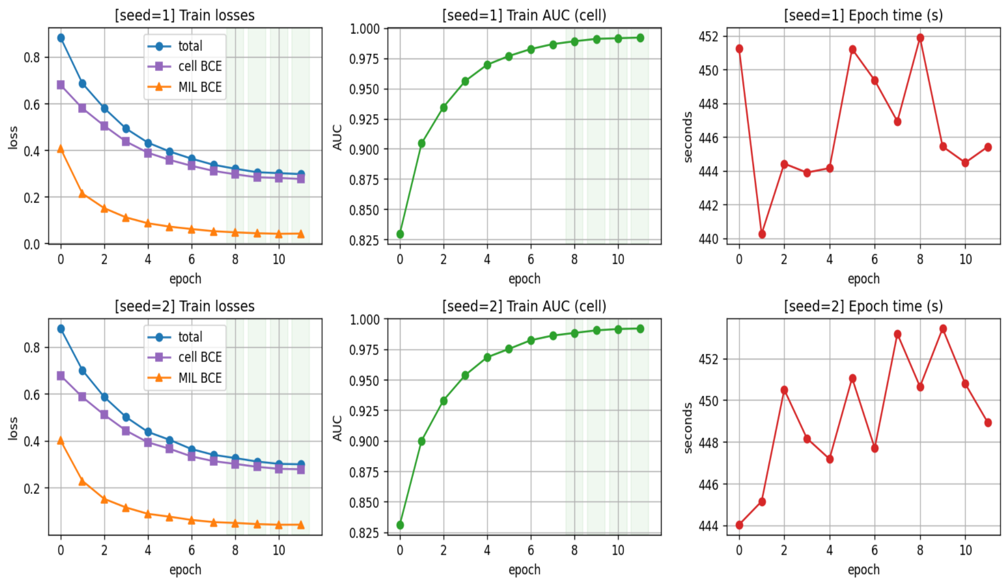

Fig. 5 v58 co-attention learning curves — train AUC + loss, 2 seeds, 12 epochs. Clean, no instability.

0.8392 ✓

intermediate CO-ATTENTION (CAFNet-style) — new best, beats v47's 0.8355

0.7823 ✗

higher resolution (192 px) — regressed

0.7376 ✗

more capacity (EfficientNetV2-S) — regressed

Of every architecture tried, only fusion design moved the needle — the model is data-bound. Capacity and resolution both hurt.

Five things I'll carry forward

01

Higher CV ≠ higher LB.

v15 had my best CV (0.866) and my worst LB (0.572). On patient-grouped OOD, CV alone is not a signal.

02

Don't stack unvalidated changes.

v43 added four tweaks at once → −0.012 LB with no attribution. One or two related variables per experiment.

03

Seed variance is huge.

Same recipe, different seed: 0.03 LB shift (v19 vs v23). Single-seed A/B claims under 0.03 are noise.

04

On small-N, data is the lever — not architecture.

Four regularisers gained +0.011; soft-pseudo distillation gained +0.042. Even the SOTA co-attention fusion ≈ plain late fusion (0.97 corr, both 0.83). Only co-attention fusion nudged architecture (+0.004 to a new best 0.8392); data was ~10x bigger.

05

Negative results are the methodology.

The v46/v47 winning recipe traces directly to six diagnosed failures. Reading the failures faithfully is what produced the win.

Code & LB history · github.com/rafallex/multimodal-microscopy-distillation

Thank you. Questions?